

Simultaneous Group-Envelope Bounds for Γ -Robust Multiple-Choice Knapsack Problems

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Abstract

Budgeted uncertainty reduces a binary optimization problem to nominal subproblems indexed by deviation thresholds, but processing every threshold relaxation can dominate computation. For a robust multiple-choice knapsack problem, exactly-one groups permit simultaneous evaluation. Dualizing a fixed-threshold capacity constraint cancels threshold-dependent baselines, leaving the maximum of a constant and a common-slope line within each local deviation interval. After sorting, prefix and suffix maxima with range accumulation evaluate one multiplier at every threshold in time linear in the threshold count and nearly linear in the option count. The resulting minimax interval bound is valid by weak duality, exact on feasible singleton intervals, and provably no weaker than the bounded-threshold group-clique relaxation. In a protocol-fixed 60-instance comparison, both certificate methods reach the prescribed tolerance; the compressed method wins every timing and achieves a 2.46-fold geometric-mean speedup, with a 95 percent design-stratified bootstrap interval from 2.29 to 2.65. Separate validation, robustness, stress, and application-derived panels support correctness and increasing separation through 5,760 groups.

Keywords: robust optimization; multiple-choice knapsack; parametric Lagrangian bounds

Area of review: Optimization.

Subject classifications: Programming, integer: robust multiple-choice knapsack; Programming, linear: Lagrangian upper bounds; Analysis of algorithms: parametric evaluation.

1 Introduction

Cardinality-budget uncertainty reduces a robust binary problem to finitely many nominal problems indexed by a scalar protection threshold (Bertsimas and Sim, 2003, 2004). The reduction is polynomial, but a conventional method may still solve one nominal relaxation for every distinct deviation. Strong bounded-threshold formulations and divide-and-conquer reduce this burden, yet interval relaxation solves can dominate on large instances (Álvarez-Miranda et al., 2013; Hansknecht et al., 2018; Büsing et al., 2023).

Many operational decisions have an exactly-one menu structure: one price per product, one configuration per component, or one service level per customer segment. Every fixed-threshold relaxation of the resulting robust model is a multiple-choice knapsack linear program. Although a single such relaxation is classical and inexpensive, jointly bounding the full parameterized family is a different computational task.

This paper identifies a cancellation created by the exactly-one equations. After dualizing a fixed-threshold capacity constraint, the threshold-dependent group baselines disappear. Within each local deviation interval, a group contributes the larger of a constant and a line whose slope is common to all active options. Prefix and suffix maxima together with range accumulation therefore evaluate one multiplier simultaneously over

the full global threshold grid. The resulting algorithm is linear in the threshold count and nearly linear in the option count after sorting.

The paper makes four contributions. First, it derives the cancellation and a two-envelope algorithm that evaluates a multiplier over all thresholds at once. Second, it turns this evaluator into a valid minimax certificate and proves that the exact certificate is no weaker than the closest bounded-threshold group-clique relaxation. Third, it embeds the certificate in an adaptive decomposition with explicit global-bound invariants and a separately scoped exact-integer extension. Fourth, a protocol-fixed study distinguishes algebraic correctness, kernel scaling, common-search-tree behavior, end-to-end certificate time, robustness, and application-derived coefficient realism. The contribution is the simultaneous parametric bound, not a new uncertainty set, a faster algorithm for one knapsack relaxation, or universal superiority of an integer solver.

2 Robust MCKP and prior-art boundary

2.1 The Γ -robust MCKP

Let $i \in \{1, \dots, n\}$ index groups and let J_i be the finite option set of group i . Option (i, j) has value $v_{ij} \in \mathbb{R}$, nominal resource contribution $a_{ij} \in \mathbb{R}$, and nonnegative deviation $d_{ij} \geq 0$. Exactly one option is selected from every group. With an integer uncertainty budget $\Gamma \in \{0, \dots, n\}$, the problem is

$$\begin{aligned} \max_x \quad & \sum_i \sum_{j \in J_i} v_{ij} x_{ij} \\ \text{s.t.} \quad & \sum_i \sum_{j \in J_i} a_{ij} x_{ij} - \max_{\substack{0 \leq u_i \leq 1 \\ \sum_i u_i \leq \Gamma}} \sum_i u_i \sum_{j \in J_i} d_{ij} x_{ij} \geq 0, \\ & \sum_{j \in J_i} x_{ij} = 1 \quad (i = 1, \dots, n), \\ & x_{ij} \in \{0, 1\}. \end{aligned} \tag{R-MCKP}$$

For integer Γ , the inner maximum is the sum of the Γ largest selected deviations. The model covers a robust resource or residual constraint; the sign convention lets larger a_{ij} represent more nominal slack.

For a fixed binary selection, write $d_i := \sum_j d_{ij} x_{ij}$ for its selected deviation in group i . LP duality for the inner budget problem gives

$$\max_{0 \leq u \leq 1, \mathbf{1}^\top u \leq \Gamma} \sum_i d_i u_i = \min_{\theta \geq 0} \left\{ \Gamma \theta + \sum_i [d_i - \theta]_+ \right\}. \tag{1}$$

Define the global candidate set

$$\mathcal{B} := \{0\} \cup \{d_{ij} : i = 1, \dots, n, j \in J_i\} = \{b_0 < \dots < b_{B-1}\}, \quad B := |\mathcal{B}|.$$

Because the right-hand side of (1) is convex piecewise linear with breakpoints at the selected deviations, a binary selection is robust feasible if and only if there is a $\theta \in \mathcal{B}$ such that

$$\sum_i \sum_{j \in J_i} r_{ij}(\theta) x_{ij} \geq \Gamma \theta, \quad r_{ij}(\theta) := a_{ij} - [d_{ij} - \theta]_+. \tag{2}$$

Thus (R-MCKP) is a finite union of fixed-threshold MCKPs. This classical representation is the starting point.

For completeness, threshold sufficiency follows directly from convexity. For a fixed selected deviation vector q , the protection function $h(\theta) = \Gamma \theta + \sum_i [q_i - \theta]_+$ is convex and piecewise linear. Away from a selected deviation its slope is $\Gamma - |\{i : q_i > \theta\}|$, so a minimum is attained at zero or at a selected deviation, with a flat interval allowed. The global set $\mathcal{B}$ contains all such candidates.

Table 1. Boundary between the present task and related algorithmic streams.

Stream	Established operation	Treatment here
Bertsimas–Sim and filtered enumeration	Solve a finite, possibly reduced set of nominal threshold problems	Simultaneously evaluates a Lagrangian multiplier over all thresholds
Fast fixed MCKP LP	Solves one MCKP relaxation using hull or partition structure	Amortizes evaluation across a parameterized family; does not improve the one-LP complexity
Divide-and-conquer	Avoids threshold solves using monotonicity	Replaces an interval LP by a direct exactly-one group-envelope bound
Strong and bounded-threshold formulations	Strengthen or linearize the robust model over a threshold interval	Proves objective-bound dominance over the group-clique interval LP without introducing a new robust formulation
Ellipsoidal robust MMKP (Caserta and Voß, 2019)	Treats multiple resources and covariance-based uncertainty	Different uncertainty set, geometry, and computational target

2.2 Finite-menu pricing specialization

Finite-menu pricing supplies one operational specialization. Product i has reference price $\bar{p}_i > 0$, exposure $w_i \geq 0$, and admissible prices $p_{ij} > 0$, $j \in J_i$; admissibility can encode the reference band $p_{ij} \in [(1 - \sigma_i)\bar{p}_i, (1 + \sigma_i)\bar{p}_i]$ for $\sigma_i \in [0, 1)$. Binary x_{ij} selects exactly one price. At option j , demand is $\tilde{g}_{ij} = \hat{g}_{ij} + \xi_i \delta_{ij}$, where $0 \leq \delta_{ij} \leq \hat{g}_{ij}$, $|\xi_i| \leq 1$, and $\sum_i |\xi_i| \leq \Gamma$. Let $\tilde{N}(x, \xi) := \sum_{i,j} w_i p_{ij} \tilde{g}_{ij} x_{ij}$ and $\tilde{D}(x, \xi) := \sum_{i,j} w_i \bar{p}_i \tilde{g}_{ij} x_{ij}$. Nominal revenue $\tilde{N}(x, 0)$ is maximized subject to $\tilde{N}(x, \xi) / \tilde{D}(x, \xi) \geq \Delta$ for every admissible ξ .

If $\tilde{D} > 0$ throughout, cross-multiplication and adverse signs subtract the Γ largest absolute selected residual deviations. The coefficients of (R-MCKP) are

$$v_{ij} = w_i p_{ij} \hat{g}_{ij}, \quad a_{ij} = w_i (p_{ij} - \Delta \bar{p}_i) \hat{g}_{ij}, \quad d_{ij} = |w_i (p_{ij} - \Delta \bar{p}_i) \delta_{ij}|. \quad (3)$$

Without positivity, only the additive residual interpretation remains. This specialization motivates the application-derived panel but is not required by the algorithm.

2.3 Prior-art boundary

Table 1 separates the task studied here from its ingredients: threshold enumeration and filtering (Bertsimas and Sim, 2003; Álvarez-Miranda et al., 2013; Lee and Kwon, 2014), divide-and-conquer (Hansknecht et al., 2018), robust formulations and inequalities (Atamtürk, 2006; Fischetti and Monaci, 2012; Monaci et al., 2013; Joung and Park, 2021; Joung et al., 2023), and fast fixed-MCKP LP algorithms (Zemel, 1980; Dyer, 1984; Szkaliczki, 2025). The closest comparator is the bounded-threshold framework with clique aggregation of Büsing et al. (2023). None of these sources, to our knowledge, evaluates one multiplier over the entire robust MCKP threshold set through exactly-one group envelopes.

The argument proceeds in three logical steps. The threshold representation first converts the robust model into a finite family of fixed-threshold linear relaxations. Lagrangian duality then replaces that family by a minimax interval certificate and permits comparison with the group-clique relaxation. Finally, the exactly-one equations expose the cancellation that makes simultaneous evaluation computationally useful. Keeping these steps separate is important: certificate validity follows from duality, whereas speed follows from the group-envelope data structure.

3 A Lagrangian bound over threshold intervals

Fix $\theta \in \mathcal{B}$ and define the group baseline

$$r_i^*(\theta) := \max_{j \in J_i} r_{ij}(\theta), \quad c_{ij}(\theta) := r_i^*(\theta) - r_{ij}(\theta) \geq 0, \quad C(\theta) := \sum_i r_i^*(\theta) - \Gamma\theta.$$

Using the exactly-one equations, (2) is equivalent to $\sum_{i,j} c_{ij}(\theta)x_{ij} \leq C(\theta)$. If $C(\theta) < 0$, the fixed-threshold problem is infeasible. If $C(\theta) \geq 0$, choosing a maximizer of $r_{ij}(\theta)$ in every group gives zero total transformed cost, so the disjunct is feasible. Its LP relaxation value is

$$L(\theta) := \max \left\{ \sum_{i,j} v_{ij}x_{ij} : \sum_{i,j} c_{ij}(\theta)x_{ij} \leq C(\theta), \quad \sum_j x_{ij} = 1, x \geq 0 \right\}. \quad (4)$$

Define the feasible-threshold index set $\mathcal{F} := \{k \in \{0, \dots, B-1\} : C(b_k) \geq 0\}$ and assume $\mathcal{F} \neq \emptyset$; if it is empty, every threshold disjunct is infeasible. The target certificate is

$$M := \max_{k \in \mathcal{F}} L(b_k), \quad (5)$$

an upper bound on the integer robust optimum obtained by relaxing every disjunct. More explicitly, if Z_k^{IP} is the integer optimum of threshold disjunct k , then the classical union representation gives $Z^{\text{robust}} = \max_{k \in \mathcal{F}} Z_k^{\text{IP}} \leq \max_{k \in \mathcal{F}} L(b_k) = M$.

Dualize the capacity constraint of (4) with $\lambda \geq 0$:

$$D(\lambda, \theta) := \lambda C(\theta) + \sum_i \max_{j \in J_i} \{v_{ij} - \lambda c_{ij}(\theta)\}. \quad (6)$$

For a contiguous index interval $I = \{\ell, \ell+1, \dots, u\} \subseteq \{0, \dots, B-1\}$, set

$$U(I) := \inf_{\lambda \geq 0} \max_{k \in I \cap \mathcal{F}} D(\lambda, b_k), \quad (7)$$

where $\max \emptyset := -\infty$. If interval I has a nonempty finite evaluated multiplier set $\Lambda_I \subseteq [0, \infty)$, define $\widehat{U}_{\Lambda_I}(I) := \min_{\lambda \in \Lambda_I} \max_{k \in I \cap \mathcal{F}} D(\lambda, b_k)$.

Proposition 1 (Validity, finite-search safety, and singleton exactness). *For every interval I ,*

$$\max_{k \in I \cap \mathcal{F}} L(b_k) \leq U(I) \leq \widehat{U}_{\Lambda_I}(I).$$

For a feasible singleton $I = \{k\}$, $U(\{k\}) = L(b_k)$.

Proof. If $I \cap \mathcal{F} = \emptyset$, then every displayed quantity equals $-\infty$ by convention, so the claim is immediate. Assume henceforth that $I \cap \mathcal{F} \neq \emptyset$. For every $k \in \mathcal{F}$ and $\lambda \geq 0$, weak duality gives $L(b_k) \leq D(\lambda, b_k)$. Maximizing over $k \in I \cap \mathcal{F}$ and then taking the infimum over λ gives the first inequality. Restricting an infimum to Λ_I can only increase its value, giving the second. On a feasible singleton, (6) is the Lagrangian dual obtained after optimizing over the product of group simplices. The feasible bounded LP (4) satisfies strong duality, hence $L(b_k) = \inf_{\lambda \geq 0} D(\lambda, b_k)$. $\square$

To compare the minimax bound formally with the closest prior-art relaxation, let $\underline{\theta} = b_\ell$, $\bar{\theta} = b_u$, and define $Q(I)$ as the optimum of the bounded-threshold group-clique LP

$$\begin{aligned}
& \max_{x,p,z} \quad \sum_{i,j} v_{ij} x_{ij} \\
& \text{s.t.} \quad \sum_{i,j} \left[-a_{ij} + [d_{ij} - \bar{\theta}]_+ \right] x_{ij} + \sum_i p_i + \Gamma z \leq -\Gamma \underline{\theta}, \\
& \quad \sum_j [\min\{d_{ij}, \bar{\theta}\} - \underline{\theta}]_+ x_{ij} \leq p_i + z \quad (i = 1, \dots, n), \\
& \quad \sum_j x_{ij} = 1 \quad (i = 1, \dots, n), \\
& \quad x, p \geq 0, \quad 0 \leq z \leq \bar{\theta} - \underline{\theta}.
\end{aligned} \tag{8}$$

This is the exactly-one specialization of the bounded-threshold clique construction of [Büsing et al. \(2023\)](#). We use the extended-value convention $Q(I) = -\infty$ if this maximization problem is infeasible.

Theorem 2 (Dominance by the exact minimax envelope). *For every threshold interval I , the exact minimax group-envelope bound is no larger than the bounded-threshold group-clique bound:*

$$\max_{k \in I \cap \mathcal{F}} L(b_k) \leq U(I) \leq Q(I).$$

Thus $U(I)$ is at least as strong as the comparator relaxation for the objective under study, while avoiding an interval LP solve.

Proof. If $I \cap \mathcal{F} = \emptyset$, then $U(I) = -\infty$ and both inequalities are immediate. Hence assume $I \cap \mathcal{F} \neq \emptyset$. The epigraph formulation of (7) is a linear program. Its dual has variables $\mu_k \geq 0$ and $y_{kij} \geq 0$ and can be written

$$\begin{aligned}
& \max \quad \sum_{k \in I \cap \mathcal{F}} \sum_{i,j} v_{ij} y_{kij} \\
& \text{s.t.} \quad \sum_{k \in I \cap \mathcal{F}} \mu_k = 1, \quad \sum_j y_{kij} = \mu_k \quad (k, i), \\
& \quad \sum_{k,i,j} c_{ij}(b_k) y_{kij} \leq \sum_k \mu_k C(b_k).
\end{aligned}$$

Strong duality applies because the primal epigraph is feasible and has a finite optimum. The dual can be read as a convex mixture of threshold-indexed group-simplex points coupled by one aggregate capacity inequality. Take any feasible dual solution and set

$$\begin{aligned}
x_{ij} &:= \sum_k y_{kij}, & z &:= \sum_k \mu_k (b_k - \underline{\theta}), \\
p_i &:= \sum_{k,j} [\min\{d_{ij}, \bar{\theta}\} - b_k]_+ y_{kij}.
\end{aligned}$$

The simplex equations and bounds on z follow immediately. For every $b_k \in [\underline{\theta}, \bar{\theta}]$,

$$[\min\{d_{ij}, \bar{\theta}\} - \underline{\theta}]_+ \leq (b_k - \underline{\theta}) + [\min\{d_{ij}, \bar{\theta}\} - b_k]_+,$$

so the group-clique rows hold after multiplication by y_{kij} and summation. Expanding the aggregate capacity row of the dual and using $\sum_j y_{kij} = \mu_k$ cancels the group baselines and gives

$$\sum_{k,i,j} r_{ij}(b_k) y_{kij} \geq \Gamma \sum_k \mu_k b_k.$$

The identity $[d - b_k]_+ = [d - \bar{\theta}]_+ + [\min\{d, \bar{\theta}\} - b_k]_+$ then gives the first constraint of (8). The mapped point has objective $\sum_{k,i,j} v_{ij} y_{kij}$, so every feasible dual value is attained by a feasible clique-LP point. Maximizing and applying strong duality proves $U(I) \leq Q(I)$. $\square$

The finite evaluated bound $\widehat{U}_{\Lambda_I}(I)$ remains valid by Proposition 1, but a deliberately coarse multiplier set need not inherit Theorem 2. The implementation combines a fixed logarithmic grid with bounded scalar refinement; the common-trace experiment separately checks its realized relationship to $Q(I)$.

The order of maximum and infimum matters. Generally $\max_k \inf_{\lambda} D(\lambda, b_k) \leq \inf_{\lambda} \max_k D(\lambda, b_k)$, so a nonsingleton interval bound can exceed the complete fixed-threshold scan. Successive splitting tightens this relaxation, and complete refinement to feasible singletons eliminates it; the evaluation algorithm below removes its computational bottleneck.

4 Simultaneous group-envelope evaluation

Theorem 3 (Baseline cancellation). *For every $\lambda \geq 0$ and $\theta \in \mathcal{B}$,*

$$D(\lambda, \theta) = \sum_i \max_{j \in J_i} \{v_{ij} + \lambda r_{ij}(\theta)\} - \lambda \Gamma \theta. \quad (9)$$

Proof. Substitute $C(\theta) = \sum_i r_i^*(\theta) - \Gamma \theta$ and $c_{ij}(\theta) = r_i^*(\theta) - r_{ij}(\theta)$ into (6). For each group,

$$\lambda r_i^*(\theta) + \max_j \{v_{ij} - \lambda[r_i^*(\theta) - r_{ij}(\theta)]\} = \max_j \{v_{ij} + \lambda r_{ij}(\theta)\}.$$

Summing and retaining $-\lambda \Gamma \theta$ proves the identity. $\square$

The cancellation is consequential because the remaining threshold dependence has only two forms. For a group i , define

$$A_i(\lambda, \theta) := \max_{j: d_{ij} \leq \theta} \{v_{ij} + \lambda a_{ij}\}, \quad (10)$$

$$Q_i(\lambda, \theta) := \max_{j: d_{ij} > \theta} \{v_{ij} + \lambda(a_{ij} - d_{ij})\}, \quad (11)$$

where the maximum of an empty set is $-\infty$.

Lemma 4 (Two-envelope representation). *For $\lambda > 0$,*

$$\max_j \{v_{ij} + \lambda r_{ij}(\theta)\} = \max\{A_i(\lambda, \theta), Q_i(\lambda, \theta) + \lambda \theta\}. \quad (12)$$

Between consecutive distinct deviations of group i , A_i and Q_i are constant. Hence the group contribution is the maximum of one constant and one line of slope λ on that local interval, with at most one crossover.

Proof. If $d_{ij} \leq \theta$, then $r_{ij}(\theta) = a_{ij}$ and the option score is constant in θ . If $d_{ij} > \theta$, then $r_{ij}(\theta) = a_{ij} - d_{ij} + \theta$, and the score has intercept $v_{ij} + \lambda(a_{ij} - d_{ij})$ and slope λ . Maximizing separately over the two sets yields (12). Those sets are unchanged between consecutive local deviations. $\square$

4.1 Prefix-suffix range algorithm

Sort each group by d_{ij} . Before multiplier queries, applying the same unmasked accumulation with $v_{ij} = 0$ and $\lambda = 1$ computes every $C(b_k)$ and hence $\mathcal{F}$. The global threshold indices split into maximal contiguous blocks on which the local saturated set $\{j : d_{ij} \leq b_k\}$ is unchanged; all options tied at a block's left endpoint

are saturated. At a fixed λ , prefix maxima of $v_{ij} + \lambda a_{ij}$ supply the constant values in (10), while reverse suffix maxima of $v_{ij} + \lambda(a_{ij} - d_{ij})$ supply the active-line intercepts in (11). If both envelopes are finite on a block, their crossover is $(A_i - Q_i)/\lambda$; a binary search finds the first global threshold at or above it. If one envelope is empty, the other applies on the entire block. Two difference arrays accumulate the resulting constant and linear pieces; cumulative sums recover the group total at all global thresholds. Finally subtract $\lambda \Gamma b_k$.

Algorithm 1 One simultaneous multiplier evaluation

Require: Global thresholds $b_0 < \dots < b_{B-1}$; deviation-sorted groups; feasibility mask $\mathcal{F}$; $\lambda \geq 0$

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1: if  $\lambda = 0$  then
2:   Set every entry to  $\sum_i \max_j v_{ij}$ 
3:   Mask indices outside  $\mathcal{F}$  by  $-\infty$ ; return the vector
4: end if
5: Initialize intercept and slope difference arrays of length  $B + 1$ 
6: for each group  $i$  do
7:   Compute prefix maxima of  $v_{ij} + \lambda a_{ij}$ 
8:   Compute suffix maxima of  $v_{ij} + \lambda(a_{ij} - d_{ij})$ 
9:   for each local deviation interval do
10:    Read its constant  $A$  and active-line intercept  $Q$ 
11:    if  $A = -\infty$  then
12:      Range-add  $Q + \lambda b_k$  on the interval
13:    else if  $Q = -\infty$  then
14:      Range-add  $A$  on the interval
15:    else
16:      Locate the clipped first index with  $b_k \geq (A - Q)/\lambda$ 
17:      Range-add  $A$  before that index and  $Q + \lambda b_k$  from it onward
18:    end if
19:  end for
20: end for
21: Cumulatively sum both arrays and subtract  $\lambda \Gamma b_k$ 
22: Mask indices outside  $\mathcal{F}$ ; return the  $B$  values

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Theorem 5 (Exact evaluation and complexity). *Let $K := \sum_i |J_i|$. In real arithmetic, after sorting the global thresholds and each group, Algorithm 1 returns the values in (9) on $\mathcal{F}$ and $-\infty$ elsewhere for a fixed multiplier in*

$$O(B + K \log(B + 1)) \text{ time and } O(B + K) \text{ working storage.}$$

Constructing the global threshold set and sorting the groups costs $O(K \log K)$. A direct score evaluation at every option–threshold pair takes $\Theta(BK)$ arithmetic operations; fully materializing those scores also takes $\Theta(BK)$ storage.

Proof. Lemma 4 proves exactness on every local interval, including repeated deviations and endpoints under the convention $d_{ij} \leq \theta$ for the saturated set. Each group produces at most $|J_i| + 1$ local intervals. Prefix and suffix maxima cost $O(K)$. Every local interval performs constant work and one binary search in the B global thresholds, totaling $O(K \log(B + 1))$. Moreover, setting all $v_{ij} = 0$ and $\lambda = 1$ in (9) yields $\sum_i \max_j r_{ij}(b_k) - \Gamma b_k = C(b_k)$, so the same range algorithm constructs the complete feasibility mask within the stated complexity. The cumulative sums and mask application cost $O(B)$. The sorted group arrays, local-interval records, and two difference arrays occupy $O(K + B)$ storage. $\square$

The implementation evaluates a fixed 26-point logarithmic grid of multipliers and then applies bounded scalar search around the best interior grid point. Grid traces are cached once and reused by every child interval. The theorem’s storage bound is attained with ragged sorted group arrays. The released vectorized code

instead pads them to an $n \times m_{\max}$ rectangle, so its general storage is $O(B + nm_{\max} + K)$; this reduces to $O(B + K)$ in every reported equal-menu experiment. It reports the smallest value actually evaluated, advanced by one binary64 representable number toward $+\infty$. The propositions concern real arithmetic; the implementation replaces exact feasibility by the stated numerical tolerance and is therefore a solver-tolerance certificate, not an interval-arithmetic proof. In exact arithmetic, imperfect multiplier minimization can weaken the bound but cannot invalidate it.

5 Adaptive certification

Let E be the set of thresholds whose fixed LPs have been solved, initialized so that $E \cap \mathcal{F} \neq \emptyset$; the feasibility mask supplies such an index whenever $\mathcal{F} \neq \emptyset$. Let $\mathcal{P}$ be a collection of active intervals covering every index in $\mathcal{F} \setminus E$; its intervals may also contain evaluated indices. Let U^{disc} be the maximum last valid bound of every removed interval not yet dominated by the incumbent, with $U^{\text{disc}} = -\infty$ initially. Maintain

$$\text{LB} := \max_{k \in E \cap \mathcal{F}} L(b_k), \quad \text{UB} := \max \left\{ \text{LB}, U^{\text{disc}}, \max_{I \in \mathcal{P}} \widehat{U}_{\Lambda_I}(I) \right\},$$

where a maximum over an empty collection is $-\infty$. The algorithm begins with the root interval, evaluates its endpoints and midpoint, and repeatedly bisects the active interval with largest bound. The same oracle-independent endpoint–midpoint choices are evaluated in every new nonsingleton interval for both compared methods. If the three root candidates contain no feasible threshold, the initialization requirement above calls for one additional index from the already constructed feasibility mask before any gap test. All timed comparison instances had a feasible endpoint or midpoint, so this fallback was not invoked there. An interval can be removed once its stored bound is within the requested tolerance of LB; its last valid upper bound is retained in U^{disc} until LB dominates it. Singleton thresholds are solved directly.

Proposition 6 (Certificate invariant). *At every iteration, $\text{LB} \leq M \leq \text{UB}$. If the procedure stops when*

$$\text{UB} - \text{LB} \leq \varepsilon \max\{1, |\text{LB}|\},$$

then it certifies (5) to scaled absolute–relative tolerance ε . If all feasible thresholds have been evaluated, then $\text{LB} = M$ and all interval records may be removed, setting $\text{UB} := \text{LB}$. Alternatively, if every active and retained discarded bound is already at most LB, then the maintained bounds satisfy $\text{LB} = \text{UB} = M$.

Proof. LB is the maximum over a subset of fixed-threshold LP values. Every unevaluated feasible threshold belongs to an active interval or to a removed interval whose retained bound is represented by U^{disc} ; [Proposition 1](#) therefore implies $M \leq \text{UB}$. Splitting replaces a parent by children whose union is the parent, and direct singleton evaluation transfers its exact LP value into LB. These operations preserve the inequalities. The stopping claim follows by division by $\max\{1, |\text{LB}|\}$. If every feasible threshold has been evaluated, the definition of M gives $M = \text{LB}$ and obsolete interval records can be deleted. Under the alternative bound-dominance condition, the displayed definition of UB gives $\text{UB} = \text{LB}$; in either case the stated equality follows. $\square$

The primary certificate concerns M , the complete threshold-disjunctive LP upper bound. The released implementation also provides an exact integer variant: it replaces the LP lower bound by a globally robust integer incumbent, solves a fixed-threshold MCKP branch-and-bound whenever a singleton must be resolved, and prunes a whole interval whenever its retained LP upper bound cannot improve the incumbent.

Corollary 7 (Exact integer integration). *The integer interval-search variant maintains $Z^{\text{incumbent}} \leq Z^{\text{robust}} \leq \text{UB}$. If it exhausts the active intervals or terminates with $\text{UB} - Z^{\text{incumbent}}$ within the prescribed scaled tolerance, the incumbent is globally optimal to that tolerance.*

Proof. Every fixed-threshold integer optimum is bounded by its fixed-threshold LP value, and [Proposition 1](#) bounds all such LP values in an active interval. Direct singleton solution transfers an exact integer value into the incumbent; splitting preserves interval coverage; and pruning occurs only when a valid retained upper bound cannot improve the incumbent. The invariant and stopping claim follow exactly as in [Proposition 6](#). $\square$

6 Computational study

6.1 Design and comparators

The revised protocol was serialized before the sparse-comparator confirmatory rerun; its SHA-256 digest is 20fd417c6765573b98c88b648abb8af74775e5d0f7d0796aa8a0097853f211ce. This is a prospectively locked internal protocol, not an externally registered study. All runs used one thread on an Apple M4 under macOS, Python 3.14.2, NumPy 2.4.4, and SciPy 1.17.1 with HiGHS. Source data, raw timing repetitions, environment records, summaries, and generation scripts accompany the paper.

Four structured families isolate distinct difficulties: *dense frontier* gives many competitive value–resource tradeoffs; *correlated risk* couples value gains and deviations; *near tie* produces unstable rankings; and *many breakpoints* gives essentially unique deviations, so B grows with K . The primary panel uses six options per group, $\Gamma = \lfloor \sqrt{n} \rfloor$, $n \in \{360, 720, 1440\}$, five seeds per family–size cell, and five timing repetitions per method. Method order alternates within an instance; the instance median is the sole statistical observation. We report geometric-mean paired speedup, a 10,000-draw design-stratified bootstrap interval that resamples seeds within each family–size cell, and an exact one-sided sign test. These summaries quantify variation across the generated instances in the fixed factorial design; they do not define a superpopulation of all robust MCKPs.

The end-to-end comparator is the bounded-threshold LP of [Büsing et al. \(2023\)](#), specialized with one clique/GUB inequality per exactly-one group as in (8) and assembled in sparse CSR form. Both methods use the same best-first splitting policy, fixed-threshold LP routine, endpoint–midpoint candidate rule, scaled tolerance $\varepsilon = 10^{-6}$, and time limit including preprocessing. They differ only in the interval upper bound: the comparator solves a bounded-threshold clique LP, whereas the compressed method evaluates the group-envelope Lagrangian bound.

The shared fixed-threshold routine constructs each group’s upper value–cost hull and greedily merges its slopes, avoiding branch-and-bound data that neither certificate needs. Held-out validation compares this LP-only path with an independent reference routine at every threshold; both certificates therefore use the same fixed-MCKP solver.

Protocol-fixed gates required maximum algebraic error at most 2×10^{-6} ; kernel speedup at least $3\times$ for $n \geq 360$; 100% primary tolerance completion; geometric-mean primary speedup at least $2\times$ with interval lower endpoint above $1.5\times$; at least 80% timing wins; root dominance in at least 95% of cases; a median win in all four families; and a median win in every robustness configuration.

6.2 Correctness validation and oracle ablations

Forty held-out irregular instances use unequal menu sizes, repeated and near-repeated deviations, signed objectives, and Γ values from 0 to n . We compare the compressed trace with (i) the dense baseline-cost trace, (ii) direct evaluation of (9), (iii) independently optimized interval bounds, (iv) the independent fixed-MCKP LP routine at every threshold, and (v) a complete fixed-threshold LP scan. The maximum absolute discrepancy is 6.58×10^{-8} , and the minimum root-bound slack over the full scan is 0.00. Both protocol-fixed validation gates pass.

The kernel ablation separates algebraic compression from adaptive search. It contains 48 instances, nine

Table 2. Kernel ablation against dense threshold materialization. Speedups are geometric means over four families and three seeds. Storage is the median (maximum) ratio of persistent numerical-array bytes and excludes interpreter overhead.

Groups	Cases	Query speedup	Total speedup	Storage ratio	Maximum error
90	12	5.82	11.83	2.58 (35.3)	1.7e-10
180	12	8.79	16.01	2.60 (70.5)	7.0e-10
360	12	11.66	20.23	2.61 (140.9)	2.1e-09
720	12	14.75	24.68	2.62 (281.6)	7.7e-09

Table 3. Primary end-to-end certificate comparison. Each reported time and evaluation count is the median across instances of that size after taking the within-instance median over five balanced timing repetitions. Aggregate speedup treats each instance once. C and Q denote the compressed and clique certificates.

Groups	Cases	Compressed (s)	Clique (s)	Geometric speedup	Wins	Fixed-threshold LPs C/Q	Clique interval LPs
360	20	0.088	0.156	1.75	20/20	12/12	8
720	20	0.163	0.361	2.23	20/20	12/12	8
1440	20	0.239	0.842	3.80	20/20	9/12	8

normalized multiplier queries, five repetitions, and balanced order. Table 2 reports geometric-mean query and total construction-plus-query speedups. The maximum identity error is 7.68×10^{-9} . For $n \geq 360$, total speedup is $22.34\times$ in geometric mean. In the many-breakpoints family, where B grows linearly with K , the descriptive log-log slopes are 0.92 for the compressed evaluator and 2.01 for dense materialization; the left panel of Figure 1 shows the separation. Storage denotes persistent numerical-array bytes rather than peak process memory and grows sharply only in that family.

To isolate the interval oracles from adaptive-tree effects, a common-trace ablation evaluates both bounds on the same 168 dyadic intervals from 24 independently generated instances. The compressed oracle wins 24 instance timings with a $3.27\times$ geometric-mean speedup; its bound is no larger than the clique bound on 100.0% of matched intervals. This comparison includes preprocessing and sparse LP assembly.

6.3 Primary result

Both methods reach tolerance on all 60 primary instances. The compressed method wins all 60 paired timings ($p = 8.67 \times 10^{-19}$), with geometric-mean speedup $2.46\times$ and 95% design-stratified bootstrap interval [2.29, 2.65]. It evaluates a median 40.4% of thresholds as fixed LPs. Final lower bounds differ by at most 0.00 relatively, and its root bound never exceeds the clique root bound within numerical tolerance. All 300/300 repeat blocks are wins (minimum $1.28\times$); median within-instance coefficients of variation are 0.72% and 0.61%. Across an instance’s interval solves, the clique comparator uses a median 100424 sparse-matrix nonzeros and 1.19 MiB of cumulative CSR arrays. The right panel of Figure 1 displays instance- and size-level scaling.

Every family median exceeds one, from $1.55\times$ for many breakpoints to $2.51\times$ for correlated risk, so the protocol-fixed breadth gate passes.

6.4 Robustness and stress evidence

The 36-instance robustness panel changes one dimension at $n = 720$: $\Gamma = 1$, twelve-option menus, or $\Gamma = \lfloor 0.2n \rfloor$. Every instance reaches tolerance. The compressed method wins 11 of 12 sparse-budget timings and all 24 timings in the other configurations; every configuration-level bootstrap interval remains above one. The larger fraction of fixed thresholds under the large budget partly explains its lower speedup than the

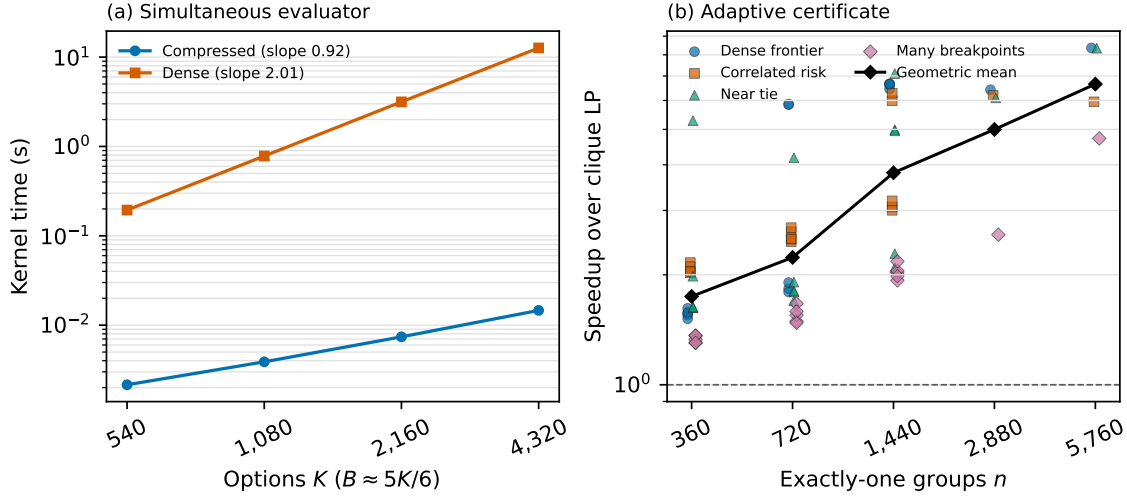


Figure 1. Mechanism and end-to-end scaling. Left: construction plus nine multiplier queries in the many-breakpoints kernel panel, where B grows linearly with K ; lines show geometric means over three seeds, with descriptive log-log slopes. Right: instance-level speedup of the compressed adaptive certificate over the clique-LP certificate; black diamonds are geometric means by size. Primary observations use five seeds at $n \leq 1440$ and stress observations use one separate seed at $n \in \{2880, 5760\}$.

Table 4. Robustness configurations at 720 groups, four families, and three new seeds. Within each configuration, bootstrap intervals resample the three seeds separately by family.

Configuration	Geometric speedup [95% interval]	Wins	Thresholds evaluated
Sparse budget ($m = 6, \Gamma = 1$)	1.88 [1.68, 2.29]	11/12	11.5%
Wide menu ($m = 12, \Gamma = \lfloor \sqrt{n} \rfloor$)	3.22 [3.01, 3.42]	12/12	23.1%
Large budget ($m = 6, \Gamma = 0.2n$)	2.98 [2.77, 3.20]	12/12	40.4%

wide-menu panel.

The stress panel is descriptive rather than inferential: it uses four families, one new seed, and two repetitions at each of 2,880 and 5,760 groups. All eight instances reach tolerance with identical final lower bounds. The geometric-mean speedup is $5.77\times$ overall and $6.65\times$ at 5,760 groups. The many-breakpoints instances contain 14,401 and 28,801 thresholds, respectively.

Table 5. Separate stress panel. Each row aggregates one new instance per family; reported times are medians across those four instances after taking the within-instance median over two balanced repetitions. Conclusions are scaling evidence, not a confidence claim.

Groups	Cases	Compressed (s)	Clique (s)	Geometric speedup	Wins
2880	4	0.344	2.151	5.00	4/4
5760	4	0.828	5.411	6.65	4/4

6.5 Exact integer integration audit

The integer integration in [Corollary 7](#) was audited separately from the confirmatory LP-certificate experiment. Twelve instances use 30 groups, all four families, three new seeds, two balanced timing repetitions, and a five-second limit. The envelope and clique interval searches use the same fixed-threshold branch-and-bound and certify 10 of 12 instances each; complete threshold enumeration certifies 11, and compact SCIP certifies all 12. Whenever at least two methods certify, their objectives differ by at most 1.14×10^{-13} . On the ten jointly certified envelope–clique cases, the geometric-mean clique/envelope time ratio is 1.00, showing that fixed-threshold integer search, rather than interval-bound computation, dominates at this scale. The many-breakpoints family remains the decisive limitation: compact SCIP certifies all three instances while each interval method certifies one. This audit validates the exact integration and its accounting invariant, but it does not support a claim of universal integer-solver superiority; the paper’s computational claim remains acceleration of the complete LP-bound certificate.

6.6 Application-derived external panel

The pricing model in [Section 2.2](#) also supplies a post-confirmatory realism check. We use the UCI Online Retail data set, which contains 541,909 transaction records ([Chen, 2015](#)). The released calibration reads the first 200,000 records; after positive-price/quantity filtering and the eight-observation SKU rule, 192,451 observations contribute to 2,549 SKU aggregates. These aggregates calibrate the segment mix, price levels, volumes, and uncertainty scales of nine semi-synthetic pricing portfolios; documented elasticity and menu-generation assumptions remain modeled rather than observed. The panel uses 360, 720, and 1,440 groups, twelve price choices, three new seeds per size, and three balanced repetitions. Both methods reach tolerance, and the compressed method wins 9/9 instances with geometric-mean speedup $1.80\times$ (95% cell-stratified bootstrap interval $[1.53, 2.19]$), rising to $2.16\times$ at 1,440 groups. This panel tests scale and coefficient realism; demand estimation is outside the algorithmic scope, and the nonrandom row cap and modeled demand curves preclude causal or transaction-level validation claims.

6.7 Scope and limitations

The experiments validate the envelope identity, retained LP certificate, and speed advantage over the tested sparse clique-LP implementation, not universal integer-solver dominance. No standard benchmark targets this relaxation, so the confirmatory generators are controlled and the external panel remains semi-synthetic; the implementation uses Python/SciPy on one architecture, and the stress sample is small. Streaming could reduce dense-oracle storage without changing its $\Theta(BK)$ arithmetic, while compiled code could shift constants. The exact audit shows that fixed-threshold integer search can erase the interval-oracle advantage and that compact SCIP remains preferable on the many-breakpoints family. DnC+, specialized fixed-MCKP solvers, and the oracle are therefore complementary.

7 Conclusion

After sorting, exactly-one group envelopes evaluate one multiplier over all thresholds in $O(B + K \log(B + 1))$ time and $O(B + K)$ working storage. The exact minimax bound is valid, singleton-exact, and no larger than the bounded-threshold group-clique LP. The protocol-fixed study finds algebraic agreement within tolerance, 60 of 60 primary wins, a $2.46\times$ geometric-mean certificate speedup, robustness across three altered regimes, and increasing size-level separation through 5,760 groups. The released exact integration preserves valid global gap accounting but also identifies the boundary of the contribution: integer subproblem work can dominate, and compact SCIP remains stronger on many-breakpoints instances. Extensions to multiple robust constraints and broader partitioned models are the next theoretical step.

Data and code availability

Code and reproducibility artifacts are available at https://github.com/eric939/robust_mckp. The evidence directory contains the serialized protocol, per-phase environments, raw CSV files, JSON summaries, and the SHA-256 artifact manifest.

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A Comparator and exact-integration implementation

The main manuscript states the bounded-threshold group-clique comparator and proves its formal relationship to the minimax bound. Both of its constraint matrices are passed to HiGHS in sparse CSR form. An ambiguous generic HiGHS status is never accepted as an upper bound: the code retries dual simplex and interior point and accepts only optimizer-certified optimality or infeasibility.

The exact integration uses the same best-first interval queue for the envelope and clique variants. A globally robust HullRound solution initializes the incumbent. The root endpoints and midpoint are solved by the fixed-threshold branch-and-bound; afterward the algorithm bisects the active interval with largest retained bound. A singleton is solved exactly only when its interval remains capable of improving the incumbent. The complete-enumeration comparator uses the same fixed-threshold branch-and-bound ordered by fixed-threshold LP value, while compact SCIP uses the standard budgeted-uncertainty reformulation. The audit contains four families, 30 groups, three seeds, two balanced repetitions, and a five-second limit. Raw results, solver statuses, node counts, interval counts, objective agreement, and environment metadata are included in the evidence directory.

B Detailed protocol and statistical estimands

The statistical unit is an instance. Timing repetitions estimate the runtime of that instance and are not treated as independent observations. For primary instance $r \in \{1, \dots, R\}$, let t_r^C and t_r^Q be the medians of five wall-clock repetitions for the compressed and clique methods. The paired speedup is $s_r = t_r^Q / t_r^C$, and the reported aggregate is

$$\exp \left(\frac{1}{R} \sum_{r=1}^R \log s_r \right).$$

The percentile interval resamples seeds with replacement separately inside each family–size cell for 10,000 draws, preserving the factorial design. It is a design-stratified measure of variability across generated seeds, not a population-sampling interval for all robust MCKPs. The exact one-sided sign test uses $\Pr\{\text{Binomial}(R, 1/2) \geq W\}$, where W is the number of instance-level wins, under the paired symmetric-null model. No timing repetition enters either inferential calculation.

The primary design has 60 instances: three sizes, four families, and five seeds. Robustness has 36 instances: three configurations, four families, and three seeds at $n = 720$. Stress has eight new instances: two sizes, four families, and one seed. The kernel has 48 instances: four sizes, four families, and three seeds. Validation has 40 independently generated irregular cases. A post-confirmatory external panel has nine UCI-calibrated semi-synthetic portfolios: three sizes and three new seeds. Thread environment variables were fixed to one for every phase.

C Exact instance generators

For auditability, we state the structured generator. Each group has a base option and $m - 1$ upgrades. Let $q_k, k = 1, \dots, m - 1$, be equally spaced from 0.8 to 14, let $w_i \sim U[8, 8.4]$, and let G be the 25-point grid from 0.2 to 3.2. The base option has $(v_{i0}, a_{i0}, d_{i0}) = (w_i, b_f, 0)$, where b_f is 4.6, 4.1, 3.9, and 4.4 for dense frontier, correlated risk, near tie, and many breakpoints, respectively. Upgrade margins are

$$a_{ik} = b_f - q_k + \eta_{ik}, \quad \eta_{ik} \sim N(0, 0.08^2).$$

Independent value noises give

$$v_{ik} = \max\{0, w_i + h_f(q_k) + \epsilon_{ik}\},$$

with $(h_f(q), \text{sd } \epsilon)$ equal to $(8\sqrt{q}, 0.35)$, $(3.15q, 0.75)$, $(2.25q, 1.65)$, and $(7.5\sqrt{q}, 0.45)$ in the same family order. With zero-based group index i , deviations are

$$\begin{aligned} d_{ik}^{\text{dense}} &= G[(7i + 3k + s) \bmod 25], \\ d_{ik}^{\text{corr}} &= \underset{g \in G}{\text{nearest}}\{0.30 + 0.19q_k + \zeta_{ik}\}, \quad \zeta_{ik} \sim N(0, 0.08^2), \\ d_{ik}^{\text{tie}} &= G[(5i + k + s) \bmod 25], \\ d_{ik}^{\text{break}} &= 0.15 + 0.003[i(m - 1) + k] + 10^{-6}s, \end{aligned}$$

where s is the reported seed. To reproduce the frozen release exactly, the NumPy seed is the first 16 hexadecimal digits of `SHA256(`v3|family|n|m|Gamma|seed`)` reduced modulo 2^{32} . The literal `v3` prefix is a legacy implementation namespace only; it does not denote a manuscript dependency or publication lineage. This seed derivation is platform-independent; the released environment record and result files remain authoritative because numerical distribution routines can vary across library versions.

The irregular validation generator independently draws 8–20 groups and 2–10 options per group, signed Gaussian values, Gaussian margins, repeated deviations from $\{0, 0.25, 0.5, 1, 2, 4, 8\}$ with occasional perturbations below 0.05, and Γ uniformly from $\{0, \dots, n\}$. These cases target algebraic edge conditions rather than timing difficulty.

For the external panel, the calibration reads the first 200,000 UCI records and aggregates positive-quantity, positive-price observations by stock code. Products with fewer than eight retained observations are removed, leaving 192,451 contributing observations and 2,549 SKU aggregates. Price–volume quantiles determine five segment shares and reference medians. Within each generated portfolio, segment-calibrated lognormal prices and volumes combine with documented elasticity ranges; twelve psychologically rounded price choices induce objective, nominal-resource, and uncertainty coefficients. The released segment aggregate has SHA-256 digest `1ba04343eb853439f184a55dbd6ac3dbeac6f6438ef1b9dcc8b0173e3a40b7f6`; the raw data are retrieved from the cited UCI record.

D Reproducibility checklist

The supplement contains the protocol and digest, phase-specific environments, instance-level and raw timing CSVs, summaries with gates, unit tests for every algebraic and certificate invariant, and a generator for all numerical manuscript artifacts and their hash manifest. Kernel storage means persistent numerical-array bytes rather than process RSS; comparator matrix storage is measured from sparse CSR arrays; the two-repetition stress panel is descriptive.

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